



THE UNIVERSITY OF TULSA

Project 1: Linear & Logistic Regression

CS-4103 Fundamentals of Machine Learning

In the Fundamentals of Machine Learning course, your projects require you to write and run Python code. You will implement or modify one or more learning algorithms, train models with multiple datasets, test your models' accuracy, and produce graphs and tables depicting your models' performances.

You will be given a code shell to start with. In the code, you will be instructed to import only certain Python libraries. Importing a library that implements an algorithm and training with a model from that library does not satisfy the requirement of implementing an algorithm. You will not need to import any libraries besides those provided. Use the code shell as a guide, but feel free to restructure the code as you prefer. You may discuss algorithms with other students or chatbots, but it is up to you to write the code and the report.

1 Overview

For this project, you will implement the linear and logistic regression algorithms. Then you will train models using seven datasets and perform a hyperparameter search to find the best learning rate and regularization parameters for each dataset.

2 Algorithms

Linear Regression: Implement two linear regression algorithms. The first algorithm you will implement is the least squares normal equation. You will also include ridge (L2) regularization. The second algorithm you will implement is gradient descent with least squares error. You will include LASSO (L1) regularization and ridge (L2) regularization. These algorithms will be used to train regression models for the regression tasks.

Logistic Regression: Implement the logistic regression algorithm. You will implement gradient descent with categorical cross-entropy loss. You will include LASSO (L1) and ridge (L2) regularization. These algorithms will be used to train classification models for classification tasks. Compute the softmax in its numerically stable form by subtracting the largest score in each row before exponentiating, which leaves the probabilities unchanged. The binary case has a numerically stable form of the cross-entropy loss as well; see the Linear & Logistic Regression slides, slide 23.

Cross Validation: Implement a K-fold cross-validation hyperparameter search algorithm that splits the provided training dataset into K equal subsets, then trains K models for each hyperparameter assignment. It should return the mean loss of the models for each hyperparameter (mean squared error for regression, mean cross-entropy loss for classification), as well as mean accuracy for classification models.

Regularization control: The gradient descent functions in the code shell take two regularization arguments. `lambda_hyperparam` sets the overall strength of the penalty, and `lasso_ridge_ratio` splits that strength between the two penalties: the LASSO (L1) term is scaled by `lasso_ridge_ratio` and the ridge (L2) term by `1 - lasso_ridge_ratio`. Set `lasso_ridge_ratio=1.0` for LASSO only and `0.0` for ridge only. This project only asks for the two pure cases.

3 Datasets for Experimentation

There are seven datasets you will use to test the models. The handwritten 4's and 9's dataset is used twice (once with linear regression and once with logistic regression), for a total of eight sets of runs. In order to demonstrate the value of regularization, you will be using a smaller subset of the data. Sample 20 datapoints without replacement for training, and test on 100 unseen datapoints. For the full digits dataset, instead train on 100 datapoints and test on 100.

The hyperparameter ranges listed for each dataset use the notation $X \in [a : b]$ to mean the evenly spaced grid $a, 2a, 3a, \dots, b$ (step size a , both endpoints included). For example, $\text{Eta} \in [0.001 : 0.02]$ means the twenty values $0.001, 0.002, \dots, 0.020$.

3.1 Regression Datasets

Synthetic regression (linear): This dataset includes points with only one feature and one target variable. The data follows a linear equation with added noise drawn from a normal distribution.

- $\text{Eta} \in [0.001 : 0.02]$
- $\text{Lambda} \in [0.01 : 0.2]$

Synthetic regression (quadratic): This dataset also includes points with only one feature and one target variable. The data follows a quadratic equation with added noise drawn from a normal distribution.

- $\text{Eta} \in [0.001 : 0.02]$
- $\text{Lambda} \in [100.0 : 2000.0]$

California Housing: The goal of this dataset from Statlib is to predict the median price of houses on a block based on some numeric features of the block, such as the median income of residents and the density of rooms per population. It includes 20640 house datapoints consisting of eight numeric attributes and one target house price.

- $\text{Eta} \in [0.01 : 0.1]$
- $\text{Lambda} \in [0.1 : 2.0]$

3.2 Classification Datasets

Synthetic classification (linear): This dataset includes only one feature and two classes. The data follows a linear decision boundary with added noise drawn from a normal distribution.

- $\text{Eta} \in [0.01 : 0.2]$
- $\text{Lambda} \in [0.05 : 1.0]$

Breast Cancer Wisconsin: This UCI dataset contains cell measurements of patients, where the goal is to predict whether a patient has breast cancer. There are 569 datapoints across 30 features.

- $\text{Eta} \in [0.1 : 2.0]$
- $\text{Lambda} \in [0.0005 : 0.01]$

Handwritten Digits: This version of the UCI ML handwritten digits dataset includes 1797 handwritten digit images. The images contain $8 \times 8 = 64$ grayscale pixels. There are 10 classes for classification, which correspond to the digits 0-9.

- **Eta** $\in [0.1 : 2.0]$
- **Lambda** $\in [0.0001 : 0.002]$

Handwritten 4's and 9's: This is a subset of the Handwritten Digits dataset for binary classification. It contains grayscale handwritten 4's and 9's. While this is a classification dataset, you will test using regression techniques to learn it alongside classification techniques.

- **Eta** $\in [0.001 : 0.02]$ for linear regression
- **Lambda** $\in [0.5 : 10.0]$ for linear regression
- **Eta** $\in [1.0 : 20.0]$ for logistic regression
- **Lambda** $\in [0.0001 : 0.002]$ for logistic regression

4 Tests

Use 10-fold cross-validation on each data set to search for the best ridge regularization hyperparameter. Then train a model on all the training data. Finally, test the accuracy of the model on the test data. Below is more detail on how you will run your linear and logistic regression models.

4.1 Linear Regression

Finding learning rate: Train the linear regression models with the gradient descent algorithm on each regression dataset without regularization (**lambda**=0). Perform a hyperparameter search to find the best value for the learning rate (**eta**).

Finding regularization parameter: Using the value found for **eta**, do a **lambda** hyperparameter search for the best ridge regularization parameter for gradient descent.

Compare models: Train a linear model on the full training data, using the gradient descent algorithm with no regularization (**lambda**=0), with ridge (L2) regularization, and with LASSO (L1) regularization. Also train a model using the normal equation algorithm with ridge regularization. Evaluate all four models on the held-out test data and compare their test mean squared errors.

Using linear regression on digits: You will also test the effectiveness of linear regression on the 4's and 9's handwritten digits classification task. In addition to providing the error of the model, find and report the accuracy of the model by rounding the predictions to 0 or 1 and treating the result as a classification.

4.2 Logistic Regression

Finding learning rate: Train the logistic regression model with the gradient descent algorithm on each categorical dataset without regularization (**lambda**=0). Perform a hyperparameter search to find the best value for the learning rate (**eta**).

Finding regularization parameter: Using the value found for **eta**, do a **lambda** hyperparameter search for the best ridge regularization parameter for gradient descent.

Compare models: Train a logistic regression model on the full training data, using the gradient descent algorithm with no regularization ($\lambda=0$), with ridge (L2) regularization, and with LASSO (L1) regularization. Evaluate all three models on the held-out test data and compare their test accuracies and test cross-entropy losses.

Compare accuracy with linear regression: Logistic regression is designed for learning classification tasks. Compare the logistic regression models' accuracy on the handwritten 4's and 9's dataset with the accuracy of the linear regression model.

5 Report

In addition to submitting your code, submit a report that includes plots from your analysis. It should include the following:

5.1 Plots

Share plots for each hyperparameter search you perform. This should include:

- Gradient Descent Regression
 - Learning rate (η) vs. mean squared error for gradient descent models on all regression datasets.
 - Regularization coefficient (λ) vs. mean squared error for gradient descent models on all regression datasets using ridge (L2) regularization.
- Gradient Descent Classification
 - Learning rate (η) vs. accuracy for gradient descent models on all classification datasets.
 - Regularization coefficient (λ) vs. cross-entropy loss for gradient descent models on all classification datasets using ridge (L2) regularization.

Above each hyperparameter search plot, report the best and worst value for the hyperparameter. Then, report the mean square error or accuracy on test data of a model trained on the full training set with the best and the worst of the hyperparameters.

There are a few additional plots to provide.

- Show a single plot per synthetic regression dataset with the fitted linear predictors of all four linear models labeled and overlaid. Comment on any differences that may occur.
- Show a single plot for the synthetic classification dataset with the decision boundaries of all three logistic models labeled and overlaid. Comment on any differences that may occur.

6 Submission

Notebooks: Submit the completed notebooks with every cell run and its output visible. Cleared notebooks will be graded as though the code did not run.

Report: Submit the report as a single PDF containing the plots, tables, and comments described above.

Where: Submit both through Harvey.